UTS UNIVERSITY OF TECHNOLOGY SYDNEY

Faculty of Engineering and IT

Assignment Cover Sheet

Subject: 41277 Control Design

Assignment Number: Demonstration Practical 2 **Date Submitted:** [TODO: dd/mm/2026]

Assignment Title: State-Feedback Control

Student Name(s) and Number(s) **Tutorial Group:** 02

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Demonstration Practice 2

Student Number	Student Name	Overall Contribution Weighting From 0 to 100%
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Each member's individual mark is the group mark scaled by their contribution level. If not indicated, all students get the same mark.

Answers to the Assignment Questions:

Q1: State-Space Model (40 marks)

Q1.1) System Modelling

Definitions and assumptions. The QUBE-Servo 2 rotary pendulum comprises a motor-driven horizontal arm and a pendulum hinged at its tip. Encoders measure the two angles [3]. Let θ denote arm angle and α pendulum angle from the upward vertical, with $\alpha = 0$ upright. Use radians. Let $\boldsymbol{e}_r$ point along the arm, $\boldsymbol{e}_t$ point in the direction of increasing θ and $\boldsymbol{e}_z$ point upward. Positive α tilts the pendulum towards $-\boldsymbol{e}_t$. This explicitly fixes the coupling signs.

The arm and pendulum are approximated as rigid uniform slender rods. The pendulum centre of mass is at $l_p = L_p/2$. Viscous damping represents friction. Motor inductance, structural flexibility, nonlinear friction and pendulum longitudinal-axis inertia are neglected. The motor is direct drive. These are nominal modelling assumptions, not measurements of the particular laboratory unit.

$$x = \begin{bmatrix}\theta & \alpha & \dot{\theta} & \dot{\alpha}\end{bmatrix}^T, \qquad u = V_m, \qquad y = \begin{bmatrix}\theta & \alpha\end{bmatrix}^T. \tag{1}$$

Thus there are $n = 4$ states, $m = 1$ input and $p = 2$ outputs. Velocities are states even though only the angles are directly measured.

Motor electrical dynamics. Kirchhoff's voltage law and the motor relations give

$$V_m = R_m i_m + L_m \frac{di_m}{dt} + e_b, \tag{2}$$

$$e_b = k_m \dot{\theta}, \qquad \tau_m = k_t i_m. \tag{3}$$

Neglecting inductance,

$$i_m = \frac{V_m - k_m \dot{\theta}}{R_m}, \qquad \boxed{\tau_m = \frac{k_t}{R_m} V_m - \frac{k_t k_m}{R_m} \dot{\theta}.} \tag{4}$$

Voltage generates torque and back emf reduces the current as speed rises.

Inertias and energy. For a uniform rod, $dm = (m/L)\, dr$. The relevant inertias are

$$J_{\text{arm}} = \int_0^{L_r} r^2 \frac{m_r}{L_r}\, dr = \frac{1}{3} m_r L_r^2, \qquad J_r = J_{\text{arm}} + J_m + J_h, \tag{5}$$

$$J_p = \int_{-L_p/2}^{L_p/2} r^2 \frac{m_p}{L_p}\, dr = \frac{1}{12} m_p L_p^2. \tag{6}$$

Here J_r includes the arm, rotor and hub. The arm rod approximation excludes the separately added rotor and hub. No inertia-disc module is included. J_p is about the pendulum centre of mass; the hinge inertia is $J_p + m_p l_p^2 = \frac{1}{3} m_p L_p^2$ by the parallel-axis theorem.

The pendulum centre-of-mass position is

$$\boldsymbol{r}_p = L_r \boldsymbol{e}_r - l_p \sin\alpha\, \boldsymbol{e}_t + l_p \cos\alpha\, \boldsymbol{e}_z. \tag{7}$$

Using $\dot{\boldsymbol{e}}_r = \dot\theta \boldsymbol{e}_t$ and $\dot{\boldsymbol{e}}_t = -\dot\theta \boldsymbol{e}_r$ gives

$$\dot{\boldsymbol{r}}_p = l_p \sin\alpha\, \dot\theta\, \boldsymbol{e}_r + (L_r\dot\theta - l_p\cos\alpha\,\dot\alpha)\boldsymbol{e}_t - l_p \sin\alpha\,\dot\alpha\, \boldsymbol{e}_z, \tag{8}$$

$$\|\dot{\boldsymbol{r}}_p\|^2 = (L_r^2 + l_p^2 \sin^2\alpha)\dot\theta^2 + l_p^2\dot\alpha^2 - 2L_r l_p \cos\alpha\,\dot\theta\dot\alpha. \tag{9}$$

Neglecting longitudinal-axis inertia, pendulum rotation about its centre of mass contributes

$$\frac{1}{2} J_p(\dot\alpha^2 + \sin^2\alpha\,\dot\theta^2).$$

Therefore the total kinetic and potential energies are

$$T = \frac{1}{2} J_r \dot\theta^2 + \frac{1}{2} m_p \|\dot{\boldsymbol{r}}_p\|^2 + \frac{1}{2} J_p(\dot\alpha^2 + \sin^2\alpha\,\dot\theta^2), \tag{10}$$

$$U = m_p g l_p \cos\alpha. \tag{11}$$

Potential energy is referenced to the hinge height and is maximal upright.

Define

$$a = J_r + m_p L_r^2, \quad b = J_p + m_p l_p^2, \quad c = m_p L_r l_p, \quad h = m_p g l_p. \tag{12}$$

The Lagrangian is

$$\mathcal{L} = T - U = \frac{1}{2}(a + b\sin^2\alpha)\dot\theta^2 + \frac{1}{2} b\dot\alpha^2 - c\cos\alpha\,\dot\theta\dot\alpha - h\cos\alpha. \tag{13}$$

Euler–Lagrange equations. For $q \in \{\theta, \alpha\}$,

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot q}\right) - \frac{\partial \mathcal{L}}{\partial q} = Q_q, \qquad Q_\theta = \tau_m - D_r\dot\theta, \quad Q_\alpha = -D_p\dot\alpha. \tag{14}$$

The arm derivatives are

$$\frac{\partial \mathcal{L}}{\partial \theta} = 0, \tag{15}$$

$$\frac{\partial \mathcal{L}}{\partial \dot\theta} = (a + b\sin^2\alpha)\dot\theta - c\cos\alpha\,\dot\alpha, \tag{16}$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot\theta}\right) = (a + b\sin^2\alpha)\ddot\theta + 2b\sin\alpha\cos\alpha\,\dot\alpha\dot\theta - c\cos\alpha\,\ddot\alpha + c\sin\alpha\,\dot\alpha^2. \tag{17}$$

The pendulum derivatives are

$$\frac{\partial \mathcal{L}}{\partial \dot\alpha} = b\dot\alpha - c\cos\alpha\,\dot\theta, \tag{18}$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot\alpha}\right) = b\ddot\alpha - c\cos\alpha\,\ddot\theta + c\sin\alpha\,\dot\alpha\dot\theta, \tag{19}$$

$$\frac{\partial \mathcal{L}}{\partial \alpha} = b\sin\alpha\cos\alpha\,\dot\theta^2 + c\sin\alpha\,\dot\theta\dot\alpha + h\sin\alpha. \tag{20}$$

Define $k_v = k_t/R_m$ and $d = D_r + k_t k_m/R_m$. The resulting nonlinear mechanical equations, including motor torque, are

$$(a + b\sin^2\alpha)\ddot{\theta} - c\cos\alpha\,\ddot{\alpha} + 2b\sin\alpha\cos\alpha\,\dot{\alpha}\dot{\theta} + c\sin\alpha\,\dot{\alpha}^2 = k_v u - d\dot{\theta}, \tag{21a}$$

$$-c\cos\alpha\,\ddot{\theta} + b\ddot{\alpha} - b\sin\alpha\cos\alpha\,\dot{\theta}^2 - h\sin\alpha = -D_p\dot{\alpha}. \tag{21b}$$

Fixing the arm and neglecting friction gives $b\ddot{\alpha} = h\sin\alpha$. A small positive tilt therefore accelerates away from upright, which provides a physical check on the gravity sign.

Nonlinear state-space representation. Define the torque expressions

$$r_1(x,u) = k_v u - dx_3 - 2b\sin x_2\cos x_2\,x_4 x_3 - c\sin x_2\,x_4^2, \tag{22}$$

$$r_2(x,u) = b\sin x_2\cos x_2\,x_3^2 + h\sin x_2 - D_p x_4. \tag{23}$$

The equations of motion become

$$\underbrace{\begin{bmatrix} a + b\sin^2\alpha & -c\cos\alpha \\ -c\cos\alpha & b \end{bmatrix}}_{M(\alpha)} \begin{bmatrix} \ddot{\theta} \\ \ddot{\alpha} \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}. \tag{24}$$

The mass-matrix determinant and inverse are

$$\Delta(\alpha) = b(a + b\sin^2\alpha) - c^2\cos^2\alpha, \tag{25}$$

$$M^{-1}(\alpha) = \frac{1}{\Delta(\alpha)}\begin{bmatrix} b & c\cos\alpha \\ c\cos\alpha & a + b\sin^2\alpha \end{bmatrix}. \tag{26}$$

Hence

$$\boxed{\dot{x} = f(x,u) = \begin{bmatrix} x_3 \\ x_4 \\ (br_1 + c\cos x_2\,r_2)/\Delta(x_2) \\ (c\cos x_2\,r_1 + (a + b\sin^2 x_2)r_2)/\Delta(x_2) \end{bmatrix}}. \tag{27}$$

For the parameters in Q1.2, $ab - c^2 > 0$ and $\Delta(\alpha) = (ab - c^2) + (b^2 + c^2)\sin^2\alpha > 0$, so this idealised model is nonsingular for every pendulum angle.

Equilibrium and Jacobian linearisation. An upright stationary equilibrium is

$$x_e = \begin{bmatrix} \theta_e & 0 & 0 & 0 \end{bmatrix}^T, \qquad u_e = 0. \tag{28}$$

Any fixed θ_e is an ideal equilibrium, since no restoring torque depends on arm angle. Choose $\theta_e = 0$ for the numerical model. Define $\delta x = x - x_e$, $\delta u = u - u_e$ and $\delta y = y - y_e$. The first-order Taylor expansion is

$$\delta\dot{x} = A_c\delta x + B_c\delta u, \qquad \delta y = C_c\delta x + D_c\delta u, \tag{29}$$

with Jacobians

$$A_c = \left.\frac{\partial f}{\partial x}\right|_e, \qquad B_c = \left.\frac{\partial f}{\partial u}\right|_e. \tag{30}$$

At equilibrium, $r_1 = r_2 = 0$ and $\Delta_0 = ab - c^2$. Also,

$$\left.\frac{\partial(r_1, r_2)}{\partial x}\right|_e = \begin{bmatrix} 0 & 0 & -d & 0 \\ 0 & h & 0 & -D_p \end{bmatrix}, \qquad \left.\frac{\partial(r_1, r_2)}{\partial u}\right|_e = \begin{bmatrix} k_v \\ 0 \end{bmatrix}. \tag{31}$$

Derivatives of M^{-1} multiply the zero equilibrium torque vector. Multiplying the above derivatives by $M^{-1}(0) = \Delta_0^{-1}\begin{bmatrix} b & c \\ c & a \end{bmatrix}$ gives

$$\boxed{A_c = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & ch/\Delta_0 & -bd/\Delta_0 & -cD_p/\Delta_0 \\ 0 & ah/\Delta_0 & -cd/\Delta_0 & -aD_p/\Delta_0 \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 \\ 0 \\ bk_v/\Delta_0 \\ ck_v/\Delta_0 \end{bmatrix}.} \tag{32}$$

Since the outputs are the two angles with no direct voltage feedthrough,

$$\boxed{C_c = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \qquad D_c = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.} \tag{33}$$

This linearisation describes small deviations about upright; large-angle motion requires the nonlinear model in (27).

Q1.2) System Parameters

Table 1 lists the manufacturer-supplied parameters relevant to the pendulum configuration. The inertia-disc parameters are excluded. Motor inductance is listed to justify its neglect, rather than being retained as a model state.

Table 1: Parameters from Table 2.2 of the QUBE-Servo 2 user manual [3].

Symbol	Description	Value
R_m	Motor resistance	$8.4\,\Omega$
k_t	Torque constant	$0.042\,\mathrm{N\,m/A}$
k_m	Back-emf constant	$0.042\,\mathrm{V\,s/rad}$
J_m	Motor rotor inertia	$4.0 \times 10^{-6}\,\mathrm{kg\,m^2}$
J_h	Attachment hub inertia	$0.6 \times 10^{-6}\,\mathrm{kg\,m^2}$
L_m	Inductance (neglected)	$1.16 \times 10^{-3}\,\mathrm{H}$
m_r	Arm mass	$0.095\,\mathrm{kg}$
L_r	Arm length, motor pivot to tip	$0.085\,\mathrm{m}$
m_p	Pendulum mass	$0.024\,\mathrm{kg}$
L_p	Pendulum full length	$0.129\,\mathrm{m}$
	Encoder resolution (quadrature)	$2\pi/2048\,\mathrm{rad/count}$
	Recommended amplifier output	$\pm 10\,\mathrm{V}$

The motor electrical time constant is

$$\tau_{\mathrm{elec}} = \frac{L_m}{R_m} = \frac{1.16 \times 10^{-3}}{8.4} = 1.3810 \times 10^{-4}\,\mathrm{s}. \tag{34}$$

This is short compared with the mechanical time scales, motivating the quasi-static current approximation. Retaining inductance would introduce motor current as a fifth state.

The damping values in Table 2 are adopted from the manufacturer's student workbook, *Pendulum State-Space Model*, in-lab exercise 6 [4]. That source reports them as experimentally obtained example values; they were not identified from this group's specific laboratory unit. No new experimental identification is claimed.

Table 2: Additional constants and their provenance.

Symbol	Description	Value	Basis
D_r	Arm damping	$0.0015\,\mathrm{N\,m\,s/rad}$	Workbook example
D_p	Pendulum damping	$0.0005\,\mathrm{N\,m\,s/rad}$	Workbook example
g	Gravity	$9.81\,\mathrm{m/s^2}$	Standard approximation

The derived dimensions and inertias are

$$l_p = L_p/2 = 0.0645\,\mathrm{m}, \tag{35}$$
$$J_{\mathrm{arm}} = \tfrac{1}{3}(0.095)(0.085)^2 = 2.2879167 \times 10^{-4}\,\mathrm{kg\,m^2}, \tag{36}$$
$$J_r = J_{\mathrm{arm}} + 4.0 \times 10^{-6} + 0.6 \times 10^{-6} = 2.3339167 \times 10^{-4}\,\mathrm{kg\,m^2}, \tag{37}$$
$$J_p = \tfrac{1}{12}(0.024)(0.129)^2 = 3.3282 \times 10^{-5}\,\mathrm{kg\,m^2}. \tag{38}$$

The constants used in the matrices are

$$a = 4.0679167 \times 10^{-4}\,\mathrm{kg\,m^2}, \qquad b = 1.33128 \times 10^{-4}\,\mathrm{kg\,m^2}, \tag{39}$$
$$c = 1.3158 \times 10^{-4}\,\mathrm{kg\,m^2}, \qquad h = 0.01518588\,\mathrm{N\,m}, \tag{40}$$
$$d = 0.00171\,\mathrm{N\,m\,s/rad}, \qquad k_v = 0.005\,\mathrm{N\,m/V}, \tag{41}$$
$$\Delta_0 = 3.68420646 \times 10^{-8}\,\mathrm{kg^2\,m^4}. \tag{42}$$

Substituting into (32) gives

$$A_c \simeq \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 54.2357795 & -6.17904785 & -1.78573054 \\ 0 & 167.674898 & -6.10719845 & -5.52075014 \end{bmatrix}, \quad B_c \simeq \begin{bmatrix} 0 \\ 0 \\ 18.0673914 \\ 17.8573054 \end{bmatrix}. \tag{43}$$

The matrices C_c, D_c remain exactly those in (33). These are nominal simulation parameters. Actual mass distribution, damping, encoder calibration and unmodelled friction may produce differences from hardware measurements.

Q1.3) System Model Discretisation

Choose the sampling time

$$\boxed{T_s = 0.001\,\mathrm{s}, \qquad f_s = 1/T_s = 1000\,\mathrm{Hz}.} \tag{44}$$

This matches the value in the supplied ControLab template's `Hardware_settings.m` [5]. It is also short relative to the mechanical dynamics, since $T_s \max_i |\lambda_i(A_c)| \simeq 0.01763 \ll 1$ for this model. Hardware execution timing and achieved closed-loop performance must still be checked during implementation.

Zero-order hold (ZOH) assumes that motor voltage is held constant between controller updates. The exact discretisation of the linearised model under this assumption is

$$A = e^{A_c T_s}, \qquad B = \int_0^{T_s} e^{A_c \tau} B_c \, d\tau, \qquad C = C_c, \quad D = D_c. \tag{45}$$

The MATLAB function `c2d` evaluates these expressions without requiring an inverse of A_c. This matters because the arm-position mode makes A_c singular. The discrete-time model is

$$\delta x[k+1] = A\delta x[k] + B\delta u[k], \tag{46a}$$
$$\delta y[k] = C\delta x[k] + D\delta u[k]. \tag{46b}$$

At the chosen zero equilibrium the deviation variables equal the original variables. The calculated matrices are

$$A \simeq \begin{bmatrix} 1 & 2.70127657\times10^{-5} & 9.96918625\times10^{-4} & -8.80391044\times10^{-7} \\ 0 & 1.00008362958 & -3.04176177\times10^{-6} & 9.97274363\times10^{-4} \\ 0 & 0.05392103923 & 0.99384536883 & -0.00174835485 \\ 0 & 0.16705290505 & -0.00607175726 & 0.99458335877 \end{bmatrix}, \tag{47}$$

$$B \simeq \begin{bmatrix} 9.00986699\times10^{-6} \\ 8.89404027\times10^{-6} \\ 0.0179959975588 \\ 0.0177536761881 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{48}$$

The dimensions are $A \in \mathbb{R}^{4\times4}$, $B \in \mathbb{R}^{4\times1}$, $C \in \mathbb{R}^{2\times4}$ and $D \in \mathbb{R}^{2\times1}$. Displayed results are rounded; full precision must be retained in MATLAB for controller and observer design.

MATLAB calculation for Q1.2–Q1.3. The following code generates both models directly from the parameters.

```
%% Physical parameters (SI units)
Rm = 8.4; kt = 0.042; km = 0.042;
mr = 0.095; Lr = 0.085; mp = 0.024; Lp = 0.129;
Jm = 4e-6; Jh = 0.6e-6;
Dr = 0.0015; Dp = 0.0005; g = 9.81;

%% Inertias and intermediate constants
lp = Lp/2;
Jr = mr*Lr^2/3 + Jm + Jh;
Jp = mp*Lp^2/12;
a = Jr + mp*Lr^2;
b = Jp + mp*lp^2;
c = mp*Lr*lp;
h = mp*g*lp;
d = Dr + kt*km/Rm;
kv = kt/Rm;
Delta0 = a*b - c^2;

%% Continuous model: [theta; alpha; theta_dot; alpha_dot]
Ac = [0, 0, 1, 0;
      0, 0, 0, 1;
      0, c*h/Delta0, -b*d/Delta0, -c*Dp/Delta0;
      0, a*h/Delta0, -c*d/Delta0, -a*Dp/Delta0];
Bc = [0; 0; b*kv/Delta0; c*kv/Delta0];
Cc = [1 0 0 0; 0 1 0 0];
Dc = zeros(2,1);
sys_ct = ss(Ac,Bc,Cc,Dc);

%% Exact zero-order-hold discretisation
ts = 0.001;
sys_dt = c2d(sys_ct,ts,'zoh');
format long g
Ac, Bc, Cc, Dc
A = sys_dt.A
B = sys_dt.B
C = sys_dt.C
D = sys_dt.D
```

Q2: State-Feedback Control (SFC) (40 marks)

Q2.1) Controllability

For the four-state continuous-time model obtained in Q1, the controllability matrix is

$$\mathcal{C} = \begin{bmatrix} B_c & A_cB_c & A_c^2B_c & A_c^3B_c \end{bmatrix}. \tag{49}$$

Using the full-precision matrices in MATLAB gives

$$\mathcal{C} \approx \begin{bmatrix} 0 & 18.06739 & -143.52761 & 2228.45594 \\ 0 & 17.85731 & -208.92687 & 5024.20650 \\ 18.06739 & -143.52761 & 2228.45594 & -34072.92634 \\ 17.85731 & -208.92687 & 5024.20650 & -76378.80248 \end{bmatrix}. \tag{50}$$

The rank is

$$\operatorname{rank}(\mathcal{C}) = 4 = n. \tag{51}$$

Therefore, the nominal continuous-time system is completely controllable. Although the system has only one motor-voltage input, the coupling between the rotary arm and pendulum allows this input to influence all four states.

The controllability matrix of the discrete-time pair (A, B) was also checked and has rank four at the selected sampling period $T_s = 0.001\,\text{s}$. Consequently, all four discrete-time closed-loop poles can be assigned using state feedback.

The MATLAB calculations are:

```
CMc = ctrb(Ac,Bc);
CMd = ctrb(A,B);
rank(CMc)
rank(CMd)
```

Q2.2) State and Input References (Equilibrium Point)

Given the desired output reference $y^\star$, the steady-state references $x^\star$ and $u^\star$ satisfy

$$\begin{bmatrix} A - I & B \\ C & D \end{bmatrix} \begin{bmatrix} x^\star \\ u^\star \end{bmatrix} = \begin{bmatrix} 0 \\ y^\star \end{bmatrix}. \tag{52}$$

The desired output is a specified arm angle while maintaining the pendulum upright:

$$y^\star = \begin{bmatrix} \theta_{\text{ref}} \\ 0 \end{bmatrix}. \tag{53}$$

At a stationary equilibrium, both angular velocities and accelerations are zero. With $\alpha = 0$, substitution into the nonlinear equations of motion gives zero required motor voltage. Hence,

$$x^\star = \begin{bmatrix} \theta_{\text{ref}} \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad u^\star = 0. \tag{54}$$

These references satisfy

$$(A - I)x^\star + Bu^\star = 0, \qquad Cx^\star + Du^\star = y^\star. \tag{55}$$

For the simulation, a 5° arm-angle step is applied at $t = 1\,\mathrm{s}$. The reference is therefore

$$\theta_{\mathrm{ref}}(t) = \begin{cases} 0, & t < 1\,\mathrm{s}, \\ 5\pi/180, & t \geq 1\,\mathrm{s}. \end{cases} \tag{56}$$

After the step,

$$x^\star = \begin{bmatrix} 0.08726646 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad u^\star = 0\,\mathrm{V}. \tag{57}$$

The pendulum reference remains zero throughout. The two output references cannot be selected independently for a stationary upright equilibrium: an arbitrary nonzero pendulum tilt cannot be maintained with the arm stationary.

Q2.3) Closed-Loop Eigenvalues

The control requirements are an overshoot below $15\,\%$ and a settling time below $5\,\mathrm{s}$. The damping ratio and natural frequency follow from

$$\zeta = \frac{-\ln(OS/100)}{\sqrt{\pi^2 + \ln^2(OS/100)}}, \qquad \omega_n = \frac{4}{\zeta\, t_s}, \tag{58}$$

and the discrete-time eigenvalues from $z_i = e^{s_i T_s}$.

A 2% settling band is used when evaluating the responses.

For an ideal second-order system, the percentage overshoot is

$$OS = 100 \exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right). \tag{59}$$

An overshoot of 15% corresponds to a damping ratio of approximately $\zeta = 0.517$. To provide design margin, choose

$$\zeta = 0.6, \qquad \omega_n = 2.5\,\mathrm{rad/s}. \tag{60}$$

The dominant continuous-time poles are

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -1.5 \pm j2. \tag{61}$$

The corresponding second-order estimates are

$$OS \approx 9.478\%, \qquad t_{\mathrm{settle}} \approx \frac{4}{\zeta\omega_n} = 2.667\,\mathrm{s}. \tag{62}$$

The remaining two poles are selected at -6 and $-8\,\mathrm{s}^{-1}$, giving faster decay rates than the dominant pair. The complete continuous-time pole selection is

$$p_s = \begin{bmatrix} -1.5 + j2 & -1.5 - j2 & -6 & -8 \end{bmatrix}. \tag{63}$$

Using $T_s = 0.001\,\mathrm{s}$, the discrete-time poles are obtained from $z_i = e^{s_i T_s}$:

$$z_{1,2} \approx 0.9984991274 \pm j0.0019970009, \tag{64}$$

$$z_3 \approx 0.9940179641, \tag{65}$$

$$z_4 \approx 0.9920319148. \tag{66}$$

All selected discrete-time poles lie strictly inside the unit circle. The second-order estimates guide the initial design; the performance of the complete fourth-order system is evaluated using the Simulink responses.

Q2.4) State-Feedback Gain

For the control law The discrete-time state-feedback control law is

$$u(k) = -F\big(x(k) - x^\star(k)\big) + u^\star(k). \tag{67}$$

The feedback gain is calculated using the full-precision discrete-time matrices and the selected poles:

```
pz = exp(ps*ts);
F = place(A,B,pz);
eig(A-B*F)
```

The resulting gain is

$$F \approx \begin{bmatrix} -0.14517718 & 15.38254012 & -0.45888105 & 0.76529169 \end{bmatrix}. \tag{68}$$

Since there is one control input and four states, $F \in \mathbb{R}^{1\times 4}$.

MATLAB gives the achieved eigenvalues

$$\lambda(A - BF) \approx \left\{ \begin{array}{l} 0.9920319148, \\ 0.9940179641, \\ 0.9984991274 + j0.0019970009, \\ 0.9984991274 - j0.0019970009 \end{array} \right\}. \tag{69}$$

These agree with the desired discrete-time poles to numerical precision. Therefore, the nominal unsaturated discrete-time closed loop is asymptotically stable. The effects of input limits and nonlinear dynamics are assessed through simulation.

Q2.5) Closed-Loop Implementation

The state-feedback controller was implemented in a MATLAB Function block in Simulink with `run_observer = 0`, so that the true simulated state vector was used for feedback:

$$x = \begin{bmatrix} \theta & \alpha & \dot{\theta} & \dot{\alpha} \end{bmatrix}^T. \tag{70}$$

The controller operates at a sampling period of $T_s = 0.001\,\text{s}$. The requested voltage is limited to the recommended range of $\pm 10\,\text{V}$:

$$u_{\text{applied}}(k) = \min\big(10,\, \max\big(-10,\, -F\big(x(k) - x^\star(k)\big) + u^\star(k)\big)\big). \tag{71}$$

The same controller was applied to both the linearised and nonlinear plant models. Both simulations used zero initial plant states, a 5° arm-angle step at $t = 1\,\text{s}$, a zero pendulum-angle reference and a simulation duration of $15\,\text{s}$.

The linear plant implements $\dot{x} = A_c x + B_c u$ and $y = C_c x$, while the nonlinear plant implements the coupled equations derived in Q1. Simulation data were exported to the MATLAB workspace and plotted with axis labels, units and an arm-angle reference trace.

Figures **??** and **??** show that the arm angle approaches the 5° reference following a damped transient. The pendulum temporarily departs from upright during the manoeuvre before returning towards zero. The applied voltage also returns towards zero as the system approaches its stationary equilibrium.

The linearised and nonlinear responses are visually close. This is consistent with the small pendulum-angle excursions in this test, for which the upright linearisation provides a good approximation of the nonlinear plant.

Performance was evaluated using the exported simulation data. Arm-angle overshoot was measured relative to the commanded step amplitude. Settling time was measured from the reference step at $t = 1$ s using a band of $5^\circ \pm 0.1^\circ$, corresponding to $\pm 2\%$ of the command. Since the pendulum reference is zero, its transient is characterised by the maximum absolute angle rather than a percentage overshoot.

Table 3: Measured Simulink performance for the 5° arm-angle step.

Quantity	Linear	Nonlinear	Requirement
Overshoot (%)	**TODO**	**TODO**	< 15
Settling time (s)	**TODO**	**TODO**	< 5
Final arm error (deg)	**TODO**	**TODO**	—
Peak $\lvert\alpha\rvert$ (deg)	**TODO**	**TODO**	—
Peak $\lvert u\rvert$ (V)	**TODO**	**TODO**	≤ 10

[Insert `assets/images/fig_q25_linear.png` here]

Figure 1: Closed-loop response of the linear model under state feedback.

[TODO: Comment on Figure 1: overshoot, settling time, steady-state error and control effort versus requirements. Add zooms.**]**

Q3: Luenberger Observer Design **(20 marks)**

The controller of Q2 needs the full state $x = [\theta \quad \alpha \quad \dot{\theta} \quad \dot{\alpha}]^T$, but the QUBE-Servo 2 measures only the two angles through its encoders, $y = C_c x = [\theta \quad \alpha]^T$. The angular velocities could be obtained by differencing the encoder signals. At $T_s = 0.001$ s, however, a single encoder count ($2\pi/2048 = 0.00307$ rad $= 0.176^\circ$ [3]) would appear as a velocity step of $176^\circ/\text{s}$. Instead, a discrete-time Luenberger observer reconstructs the full state from the measured angles and the applied voltage, using the model of Q1.

Q3.1) Observability

With $n = 4$ states and $p = 2$ outputs, the observability matrix of the continuous-time model (29) is the $np \times n = 8 \times 4$ matrix

$$\mathcal{O} = \begin{bmatrix} C_c \\ C_c A_c \\ C_c A_c^2 \\ C_c A_c^3 \end{bmatrix} \approx \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \hline 0 & 54.236 & -6.179 & -1.786 \\ 0 & 167.675 & -6.107 & -5.521 \\ \hline 0 & -634.55 & 49.086 & 75.128 \\ 0 & -1256.9 & 71.453 & 209.06 \end{array}\right], \tag{72}$$

computed with the full-precision matrices of Q1:

```
OM = obsv(Ac,Cc);
```

```
rank(OM)
rank(obsv(A,C))
```

The first four rows of (72) form the 4×4 identity matrix. This is because C_c selects the two angles and $C_c A_c$ selects the two angular velocities (the first two rows of A_c in (32) are [0 0 1 0] and [0 0 0 1]). Therefore

$$\operatorname{rank}(\mathcal{O}) = 4 = n, \tag{73}$$

and the continuous-time system is **completely observable**.

From a control viewpoint, the two unmeasured states are the time derivatives of the two measured outputs. Any initial velocity therefore leaves a distinguishable trace in $y(t)$ almost immediately, so the full state can be reconstructed from a short window of input–output data. Unlike controllability in Q2.1, which relies on the arm–pendulum coupling, here every state is reached through the kinematic chain from angle to velocity alone. The discrete-time pair (A, C) also has rank 4. Sampling cannot destroy observability here because T_s is much shorter than every open-loop time constant (open-loop eigenvalues 0, 9.95, -4.02 and $-17.63\ \mathrm{s}^{-1}$).

Q3.2) Observer Eigenvalues

By the separation principle [1], the eigenvalues of the observer-based closed loop are the union of the state-feedback eigenvalues, $\lambda(A - BF)$, and the observer eigenvalues, $\lambda(A - LC)$. The two designs can therefore be carried out independently. The estimation error should decay clearly faster than the regulated response, so that by the time the controller's dynamics evolve, it is effectively acting on the true state.

The state-feedback poles from Q2.3 are $p_s = \{-1.5 \pm j2,\ -6,\ -8\}\ \mathrm{s}^{-1}$. A common design rule places the observer poles two to six times faster than the fastest controller pole. Here a factor of five is used, with distinct real values:

$$p_{o,s} = \{-40,\ -45,\ -50,\ -55\}\ \mathrm{s}^{-1}. \tag{74}$$

The reasons for this choice are:

- The slowest observer mode has time constant $1/40 = 25\ \mathrm{ms}$, so the estimation error is reduced by a factor $e^{-4} \approx 2\%$ within about 0.1 s. That is 27 times faster than the $1/1.5 = 0.67\ \mathrm{s}$ time constant of the dominant controller pair, and far shorter than the 2.7 s closed-loop settling time.
- Real poles give a non-oscillatory error decay. They are distinct because `place` only allows a pole multiplicity of up to $\operatorname{rank}(C) = 2$.
- The poles are not pushed further left. A faster observer needs a larger gain L, which passes more encoder quantisation noise into $\hat{x}$ and increases the initial "peaking" of the velocity estimates (Q3.4). With $|p_{o,s}|T_s \leq 0.055$, each observer time constant still spans at least 18 samples, so the discrete design is well resolved.

Mapping these poles to the z-plane with $z_i = e^{s_i T_s}$ and $T_s = 0.001\ \mathrm{s}$ (MATLAB: `po_z = exp(po_s*ts)`) gives

$$p_{o,z} \approx \{0.96079,\ 0.95600,\ 0.95123,\ 0.94649\}. \tag{75}$$

All lie on the positive real axis, strictly inside the unit circle and closer to the origin than the controller poles of Q2.3, $\{0.99850 \pm j0.00200,\ 0.99402,\ 0.99203\}$. This confirms in the z-domain that the estimation error decays faster than the closed-loop response.

Q3.3) Observer Gain

The discrete-time Luenberger observer is

$$\hat{x}(k+1) = A\hat{x}(k) + Bu(k) + L\big(y(k) - \hat{y}(k)\big), \qquad \hat{y}(k) = C\hat{x}(k). \tag{76}$$

It is a copy of the discrete model (46), driven by the same input and corrected by the output innovation $y(k) - \hat{y}(k)$. Subtracting it from the plant model, the estimation error $e(k) = x(k) - \hat{x}(k)$ evolves as

$$e(k+1) = (A - LC)\, e(k), \tag{77}$$

independently of $u(k)$ and of the reference. Because $\lambda(A - LC) = \lambda(A^T - C^T L^T)$, computing L is the dual of a state-feedback design for the pair (A^T, C^T). That design is solvable because (A, C) is observable (Q3.1):

```
po_s = [-40 -45 -50 -55];
po_z = exp(po_s*ts);
Lo   = place(A',C',po_z)';
eig(A - Lo*C)
```

The resulting $n \times p = 4 \times 2$ observer gain is

$$L \approx \begin{bmatrix} 0.087990 & -0.005869 \\ -0.011086 & 0.086021 \\ 1.6806 & -0.25164 \\ -0.70415 & 1.8098 \end{bmatrix}, \tag{78}$$

and MATLAB returns

$$\lambda(A - LC) \approx \{0.96079,\ 0.95600,\ 0.95123,\ 0.94649\}, \tag{79}$$

which matches (75) to numerical precision. The observer therefore has the desired eigenvalues, and the error dynamics (77) are asymptotically stable.

The structure of L has a clear physical reading. Column 1 weights the arm innovation $\theta - \hat{\theta}$, and column 2 weights the pendulum innovation $\alpha - \hat{\alpha}$. Each angle estimate is corrected mainly by its own measurement, by about 0.088 of the innovation per sample (diagonal of the first two rows). Each velocity estimate is corrected by about $1.7\ (\mathrm{rad/s})/\mathrm{rad}$ per sample from its own angle innovation, which acts as a model-based, filtered differentiator. One encoder count of innovation therefore changes $\hat{\dot{\theta}}$ by only about $1.68 \times 0.00307\ \mathrm{rad/s} = 0.30°/\mathrm{s}$, compared with the $176°/\mathrm{s}$ jump of a raw finite difference. The off-diagonal terms account for the arm–pendulum coupling in A.

Q3.4) Closed-Loop Implementation

The observer runs in the same MATLAB Function block as the state feedback of Q2.5, sampled at $T_s = 0.001$ s, with `run_observer = 1`. The controller now feeds back the estimated state:

$$u(k) = -F\big(\hat{x}(k) - x^\star(k)\big) + u^\star(k), \tag{80}$$

where F, $x^\star$ and $u^\star = 0$ are unchanged from Q2.2–Q2.4. The relevant part of the block is:

```
x_hat = x_hat_k; % estimate used at sample k
x_ref = [y_ref(1); 0; 0; 0]; % Q2.2
u_ref = 0;
u_k = -F*(x_hat_k - x_ref) + u_ref; % SFC on the estimated state
u_k = min(max(u_k, -10), 10); % +/-10 V amplifier limit
y_hat = C*x_hat_k; % Luenberger observer (Q3.3)
x_hat_k = A*x_hat_k + B*u_k + Lo*(y_k - y_hat);
```

The observer is updated with the *saturated* voltage that actually reaches the motor. If it were fed the unsaturated command, its model input would differ from the plant input whenever the amplifier limits, and $\hat{x}$ would drift.

The same controller and observer were applied to the linearised and nonlinear Simulink plants of Q2.5. As Remark 5 asks, the plant starts at rest upright, $x(0) = 0$, while the observer is deliberately initialised with

$$\hat{x}(0) = \begin{bmatrix} 3° & -1° & 0 & 0 \end{bmatrix}^T \neq x(0). \tag{81}$$

In the observer tests the 5° arm step is applied at $t = 5\,\mathrm{s}$ instead of $1\,\mathrm{s}$. The reason is explained under "Effect of the initial estimation error" below: this separates the start-up transient caused by $\hat{x}(0)$ from the tracking test. The step time is set from `main_SFC.m` at run time, and the Q2.5 configuration is unchanged. Signals were exported to the workspace and plotted with `plot_closed_loop.m`. The script `run_all_cases.m` regenerates every Q3 figure.

Observer convergence. Figure 2 compares the true and estimated states of the nonlinear model, with a zoom on the first $0.3\,\mathrm{s}$. Figure 3 shows the estimation error on a logarithmic scale. The angle estimates converge onto the measured angles within about $25\,\mathrm{ms}$, and both angle errors stay below 0.01° after $153\,\mathrm{ms}$ in both the linear and nonlinear cases. This agrees with Q3.2, since reducing a 3° error to 0.01° takes $\ln(300) \approx 5.7$ time constants of the slowest observer mode, or about $143\,\mathrm{ms}$. After the first few samples, the error curves in Figure 3 are straight lines on the logarithmic axis, which is the exponential decay predicted by (77). The final estimation error is about 10^{-13}, at the level of numerical round-off.

The zoom also shows the price of estimating unmeasured states. The velocity estimates start with zero error, but the initial 3° and -1° angle innovations pass through the velocity rows of L. They therefore produce velocity-estimate errors that peak at $41.8^\circ/\mathrm{s}$ ($\dot{\theta}$) and $28.5^\circ/\mathrm{s}$ ($\dot{\alpha}$) about $20\,\mathrm{ms}$ after start-up, before decaying with the rest of the error. This transient "peaking" grows with observer speed, and it is the main reason the observer poles were not placed further left in Q3.2.

Effect of the initial estimation error. Figures 4 and 5 show the full closed-loop responses. Because $u(k)$ is computed from $\hat{x}(k)$, the wrong initial estimate acts on the plant like an initial-condition disturbance. The controller "sees" the pendulum 1° from upright and responds with $u(0) = 0.276\,\mathrm{V}$. The gain on the pendulum angle, $F_\alpha = 15.38\,\mathrm{V/rad}$, is about 100 times the gain on the arm angle, $F_\theta = -0.145\,\mathrm{V/rad}$. This is the correct design for an inverted pendulum, but it means a small pendulum-estimate error commands a large arm motion. Over the $\approx 0.1\,\mathrm{s}$ the estimate takes to converge, the plant is pushed far enough that the arm swings to 18.3° and the pendulum to 1.61°. The recovery is then governed by the deliberately gentle dominant poles of Q2.3 ($\sigma = -1.5\,\mathrm{s}^{-1}$) and takes roughly 3 s. The peak voltage, $0.71\,\mathrm{V}$, stays well inside the $\pm 10\,\mathrm{V}$ limit, so the loop remains linear throughout. By $t = 5\,\mathrm{s}$ the arm is back within 0.01° of zero, which is why the tracking step was moved there.

Closed-loop performance. Figure 6 zooms in on the 5° step, and Table 4 compares it with the full-state feedback results of Q2.5. Once the estimate has converged, $\hat{x}(k) \approx x(k)$ and, by the separation principle, the response is governed only by $\lambda(A - BF)$. The observer-based step response is accordingly indistinguishable from Q2.5: $8.39\,\%$ overshoot against $8.37\,\%$, and $2.715\,\mathrm{s}$ settling time against $2.713\,\mathrm{s}$. Both are well within the requirements of $< 15\,\%$ and $< 5\,\mathrm{s}$, with negligible steady-state error. The estimated arm angle in Figure 6 lies on top of the true one during the whole manoeuvre.

Table 4: Simulated 5° arm-step metrics with full-state feedback (Q2.5, step at 1 s) and observer-based feedback (Q3.4, step at 5 s). Overshoot and $2\,\%$ settling time are measured from the step; peak values are taken after the step.

Model	**Feedback**	**OS [%]**	t_s **[s]**	**Final error [deg]**	$\max\lvert\alpha\rvert$ **[deg]**	$\max\lvert V_m\rvert$ **[V]**
Linear	$x(k)$	8.37	2.713	5.5×10^{-10}	0.104	0.040
Nonlinear	$x(k)$	8.37	2.713	5.5×10^{-10}	0.104	0.040
Linear	$\hat{x}(k)$	8.39	2.715	2.9×10^{-6}	0.11	0.04
Nonlinear	$\hat{x}(k)$	8.39	2.715	2.9×10^{-6}	0.11	0.04
Requirement		< 15	< 5	–	–	≤ 10

The small differences between the two rows of each pair are the last remnant of the start-up transient, which has not fully decayed at $t = 5\,\mathrm{s}$. The linear and nonlinear results agree to the precision shown, because the pendulum never leaves $|\alpha| < 1.7^\circ$. At that angle the approximation $\sin\alpha \approx \alpha$ is accurate to about $0.01\,\%$, so the linearised model on which both F and L are based describes the plant faithfully.

On the real QUBE-Servo 2, two practical points follow. First, the start-up swing can be avoided by initialising $\hat{\theta}(0)$

and $\hat{\alpha}(0)$ from the first encoder readings. Then only the velocities, which are zero at rest, need to be estimated, and the initial innovation is close to zero. Second, the simulated steady-state error is zero only because the model is exact. Unmodelled Coulomb friction and parameter mismatch on the hardware would leave an arm-angle offset that this proportional state feedback cannot remove without integral action.

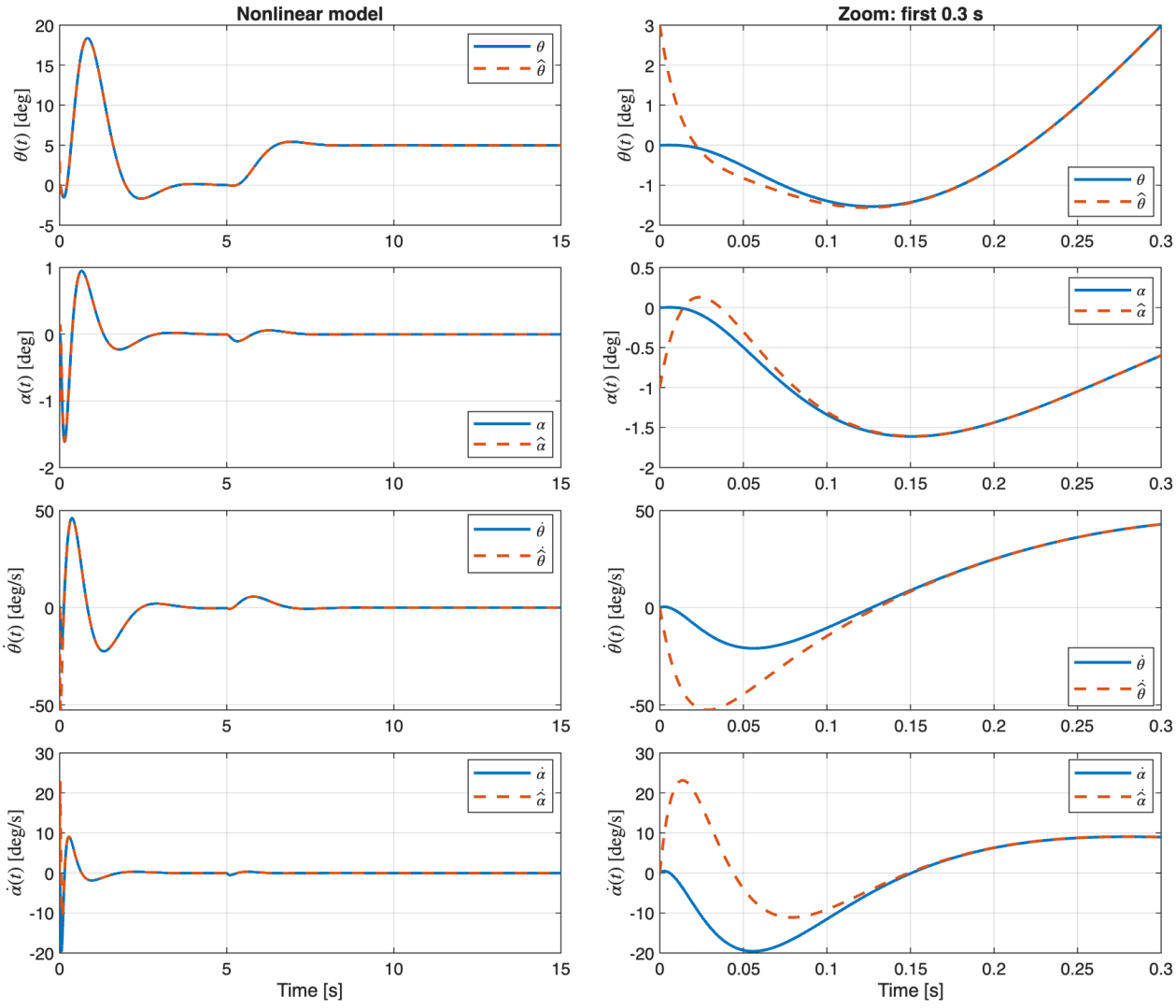


Figure 2: True (x, solid) and estimated ($\hat{x}$, dashed) states of the nonlinear model with $\hat{x}(0) = [3^\circ \ -1^\circ \ 0 \ 0]^T \neq x(0) = 0$. Left: full simulation; right: zoom on the first 0.3 s showing observer convergence.

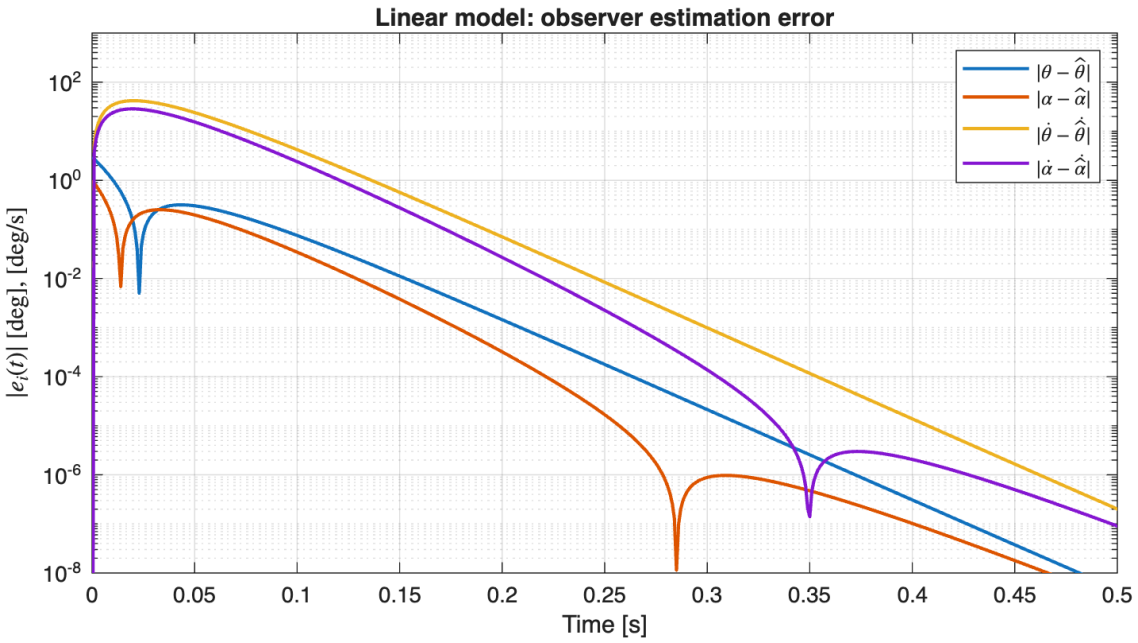


Figure 3: Magnitude of the estimation error $e = x - \hat{x}$ (linear model, logarithmic scale) during the first 0.5 s. The straight-line decay reflects the exponential error dynamics $e(k+1) = (A - LC)e(k)$.

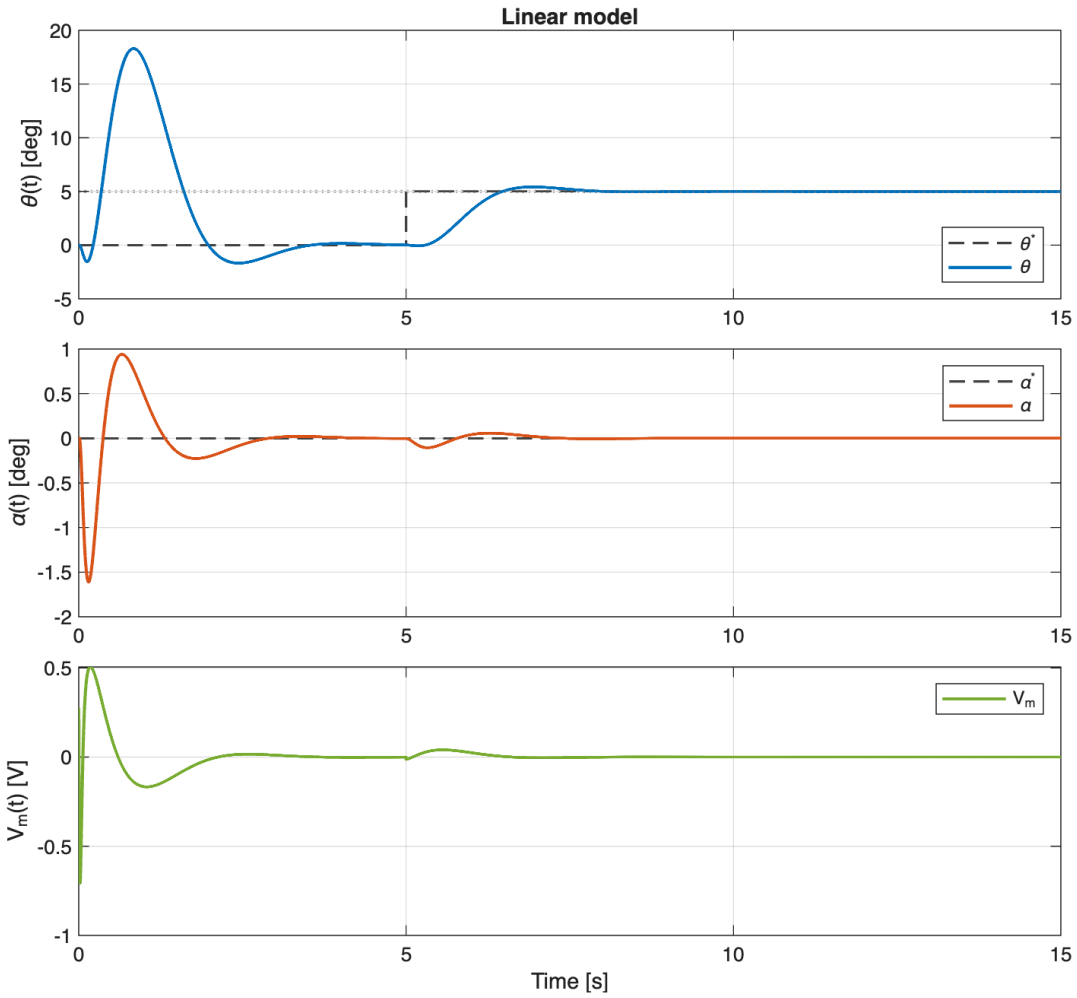


Figure 4: Observer-based state feedback on the linear model: arm angle θ with reference $\theta^\star$ (dotted lines: $\pm 2\,\%$ band), pendulum angle α and motor voltage V_m. The transient before $t = 5$ s is caused by the initial estimation error.

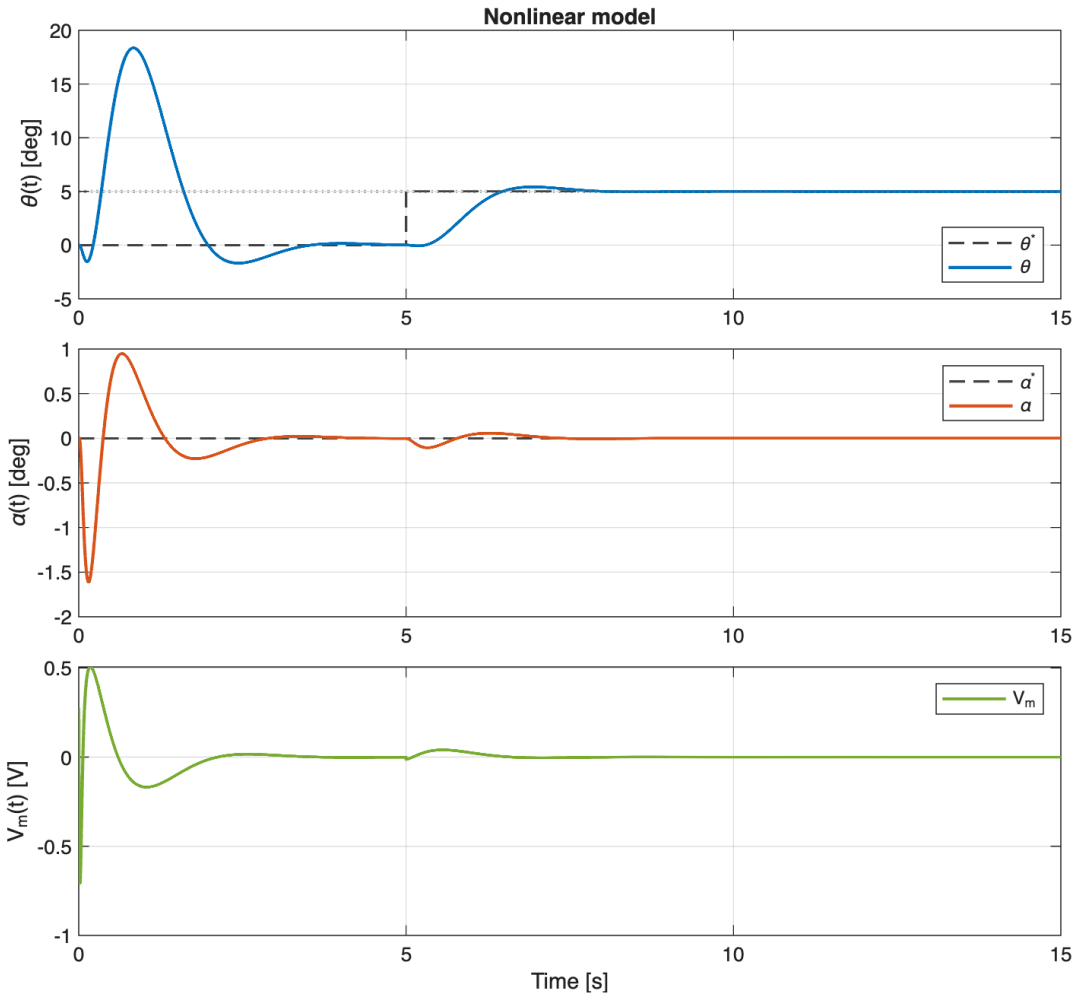


Figure 5: Observer-based state feedback on the nonlinear model: arm angle θ with reference $\theta^\star$ (dotted lines: $\pm 2\,\%$ band), pendulum angle α and motor voltage V_m.

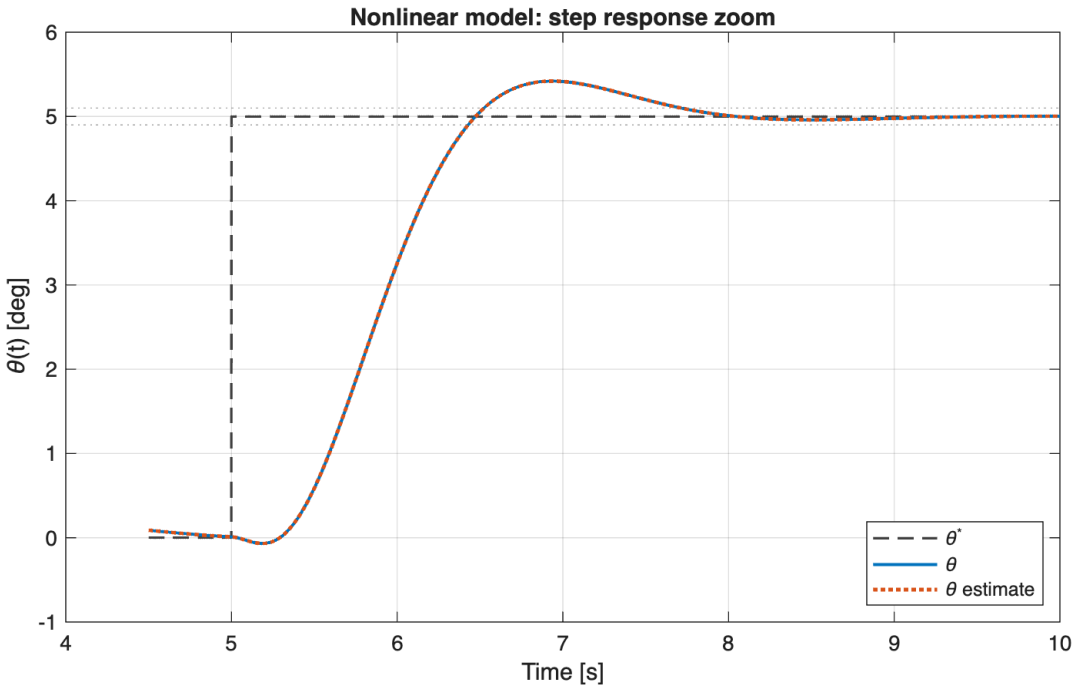


Figure 6: Zoom on the 5° arm step applied at $t = 5$ s (nonlinear model): reference, true arm angle and its estimate, with the $\pm 2\,\%$ settling band (dotted lines).

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[2] B. Messner and D. Tilbury, *Control Tutorials for MATLAB and Simulink*, University of Michigan, USA. [Online]. Available: `http://ctms.engin.umich.edu/CTMS`

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